Topic 5

Generation and transmission of electricity

5.1 Describe current as the rate of flow of charge and voltage as an electrical pressure giving a measure of the energy transferred

5.2 Define power as the energy transferred per second and measured in watts

5.3 Use the equation:

electrical power (watt, W) = current (ampere, A) × potential difference (volt, V)
$P = I \times V$

5.4 *Investigate the power consumption of low-voltage electrical items*

5.5 Discuss the advantages and disadvantages of methods of large-scale electricity production using a variety of renewable and non-renewable resources

5.6 Demonstrate an understanding of the factors that affect the size and direction of the induced current

5.7 *Investigate factors affecting the generation of electric current by induction*

5.8 Explain how to produce an electric current by the relative movement of a magnet and a coil of wire:

a on a small scale

b in the large-scale generation of electrical energy

5.9 Recall that generators supply current which alternates in direction

5.10 Explain the difference between direct and alternating current

5.11 Recall that a transformer can change the size of an alternating voltage

5.12 **Use the turns ratio equation for transformers to predict either the missing voltage or the missing number of turns**

5.13 Explain why electrical energy is transmitted at high voltages, as it improves the efficiency by reducing heat loss in transmission lines

5.14 Explain where and why step-up and step-down transformers are used in the transmission of electricity in the National Grid

5.15 Describe the hazards associated with electricity transmission

5.16 Recall that energy from the mains supply is measured in kilowatt-hours

5.17 Use the equation:

cost (p) = power (kilowatts, kW) × time (hour, h) × cost of 1 kilowatt-hour (p/kW h)

5.18 Demonstrate an understanding of the advantages of the use of low-energy appliances

5.19 Use data to compare and contrast the advantages and disadvantages of energy-saving devices

5.20 Use data to consider cost-efficiency by calculating payback times

5.21 Use the equation:

power (watt, W) = energy used (joule, J) / time taken (second, s)

$$P = \frac{E}{t}$$

Topic 6

Energy and the future

6.1 Demonstrate an understanding that energy is conserved

6.2 Describe energy transfer chains involving the following forms of energy: thermal (heat), light, electrical, sound, kinetic (movement), chemical, nuclear and potential (elastic and gravitational)

6.3 Demonstrate an understanding of how diagrams can be used to represent energy transfers

6.4 Apply the idea that efficiency is the proportion of energy transferred to useful forms to everyday situations

6.5 Use the efficiency equation:

$$\textit{efficiency} = \frac{\textit{(useful energy transferred by the device)}}{\textit{(total energy supplied to the device)}} \times 100\%$$

6.6 Demonstrate an understanding that for a system to be at a constant temperature it needs to radiate the same average power that it absorbs

6.7 *Investigate how the nature of a surface affects the amount of thermal energy radiated or absorbed*

Unit P2: Physics for the future

Overview

Content and How Science Works overview

In Unit P2 students study six topics that give them the opportunity to develop their understanding of significant concepts and relate them to important uses both for today and the future. Electricity is explained further, building on Unit P1. Students are introduced to motion, forces and momentum. Nuclear reactions and nuclear power are then discussed, including the uses and dangers of radioactivity.

Practical work in this unit will give students opportunities to plan practical ways to answer scientific questions; devise appropriate methods for the collection of numerical and other data; assess and manage risks when carrying out practical work; collect, process, analyse and interpret primary and secondary data; draw evidence-based conclusions; and evaluate methods of data collection and the quality of the resulting data.

Work on static and current electricity and on nuclear reactions provides opportunities to use models to explain ideas and processes. Students will work quantitatively when studying charge, resistance, electrical power, motion, energy, momentum and half-lives. They will have opportunities to communicate scientific information using scientific and mathematical conventions and symbols during work on resistance and motion.

Work on the applications of static electricity, car safety features, stopping distances, nuclear energy and the uses of radioactivity allow students to consider the role that physics and physicists play in providing safe and useful machines. Students also have the opportunity to consider the advantages, disadvantages and risks of these applications, and the safe uses of radioactive substances.

In Topic 1 students will learn about static electricity before discussing some uses and dangers of electrical charges. Direct current is introduced.

Topic 2 leads students to understand the relationship between current, voltage and resistance. Equations for electrical power and energy transferred are also used. Further investigations lead to an understanding of how current varies with voltage in some common components.

In Topic 3 students will develop an understanding of the motion of objects and Newton's second law of motion. This is then exemplified by considering the motion of an object as it falls through a vacuum and the atmosphere.

In Topic 4 students will learn about conservation of momentum by investigating collisions between bodies. This will enable students to apply ideas about rate of change of momentum to crumple zones, seat belts and air bags. Students will then develop an understanding of the relationship between work done, energy transferred and power.

In Topic 5 students will develop an understanding of radioactive decay, including chain reactions, and difference between fission and fusion. Students will use this context to study the role of the wider scientific community in validating theories.

In Topic 6 students will develop an understanding of the uses of different ionising radiations. They will compare and contrast the advantages and risks involved and use models to investigate radioactive decay. Students will research and discuss the advantages and disadvantages of using nuclear power for generating electricity.

Assessment overview

This unit is externally assessed, through a one hour, 60 mark, tiered written examination, containing six questions.

The examination will contain a mixture of question styles, including objective questions, short answer questions and extended writing questions.

Practical investigations in this unit

Within this unit, students will develop understanding of the process of scientific investigations, including that investigations:

- use hypotheses which are tested
- require assessment and management of risks
- require the collection, presentation, analysis and interpretation of primary and secondary evidence, including the use of appropriate technology
- should include review of methodology to assess fitness for purpose
- should include a review of hypotheses in the light of outcomes.

The following specification points are practical investigations that exemplify the scientific process and may appear in the written examination for this unit:

2.6 *Investigate the relationship between potential difference (voltage), current and resistance*

3.15 *Investigate the relationship between force, mass and acceleration*

4.3 *Investigate the forces required to slide blocks along different surfaces, with differing amounts of friction*

4.8 *Investigate how crumple zones can be used to reduce the forces in collisions*

6.8 *Investigate models which simulate radioactive decay*

The following are further suggestions for practical work within this unit:

- *Investigate forces between charges*
- *Conduct experiments to show the relationship between potential difference (voltage), current and resistance, for a component whose resistance varies with a given factor, such as temperature, light intensity and pressure*
- *Investigate the motion of falling*
- *Investigate momentum during collisions*
- *Investigate power by running up the stairs or lifting objects of different weights*

The controlled assessment task (CAT) for the GCSE in Physics will be taken from any of these practical investigations (specification points and further suggested practical investigations). This task will change every year, so future CATs will be chosen from this list.

Detailed unit content

In this specification bold text refers to higher tier only content. Italic text refers to practical investigations, which students are required to demonstrate an understanding of.

Throughout the unit

0.1 Use equations given in this unit, or in a given alternate form

0.2 **Use and rearrange equations given in this unit**

0.3 Demonstrate an understanding of which units are required in equations

Topic 1

Static and current electricity

1.1 Describe the structure of the atom, limited to the position, mass and charge of protons, neutrons and electrons

1.2 Explain how an insulator can be charged by friction, through the transfer of electrons

1.3 Explain how the material gaining electrons becomes negatively charged and the material losing electrons is left with an equal positive charge

1.4 Recall that like charges repel and unlike charges attract

1.5 Demonstrate an understanding of common electrostatic phenomena in terms of movement of electrons, including:

- a shocks from everyday objects
- b lightning
- c attraction by induction such as a charged balloon attracted to a wall and a charged comb picking up small pieces of paper

1.6 Explain how earthing removes excess charge by movement of electrons

1.7 Explain some of the uses of electrostatic charges in everyday situations, including paint and insecticide sprayers

1.8 Demonstrate an understanding of some of the dangers of electrostatic charges in everyday situations, including fuelling aircraft and tankers together with the use of earthing to prevent the build-up of charge and danger arising

1.9 Recall that an electric current is the rate of flow of charge

1.10 Recall that the current in metals is a flow of electrons

1.11 Use the equation:

charge (coulomb, C) = current (ampere, A) × time (second, s)
$Q = I \times t$

1.12 Recall that cells and batteries supply direct current (d.c.)

1.13 Demonstrate an understanding that direct current (d.c.) is movement of charge in one direction only